



An Adaptive Neuron-Proportional-Integral Controller Type Power System Stabilizer for Single-Machine Infinite-Bus System

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Abstract: In this study, an adaptive linear neuron (ADALINE) network is employed to adjust the parameters of a conventional proportional-integral (PI) controller. The proposed adaptive neuron controller (NEURON-PI) has been used here instead of the conventional power system stabilizer (PSS) to control the rotor speed deviation of the single-machine infinite-bus (SMIB) system. The rotor speed deviation is used as a control error fed to the neural network in order to adjust the PI controller gains. This strategy achieves an optimal response of the power system against a wide range of operational conditions. The linearized Heffron-Phillips model of the SMIB system with the proposed NEURON-PI controller is simulated using MATLAB R2025a/Simulink software. The simulation results have shown that the power system using the proposed control technique becomes more reliable for a wide range of operating conditions as the speed deviation and power angle deviation become less oscillatory compared to the conventional PSS controller tuned by the equilibrium optimizer algorithm (EOA) reported in a previous study.

Keywords: Dynamic stability; Power system stabilizer; Heffron-Phillips model; Single-machine infinite-bus; Adaptive linear neural network

1 Introduction

Since electrical power systems generally include mechanical, hydraulic, and electrical subsystems, they are characterized by non-linearity, uncertainty, and relatively high sensitivity to changes in operating points. Furthermore, such systems continuously suffer from the occurrence of low-frequency electromechanical vibrations caused by generator failures, short circuits, sudden load loss, and/or sudden load activation [1, 2]. Automatic voltage regulator (AVR) is one of the most important tools that has been used for decades to reduce the effects of load variations on operating voltage in power systems. With the increasing complexity of electrical power systems that accompanies the growing demand for electrical energy, the challenges of controlling such systems have become correspondingly more serious [3, 4]. To improve momentary stability and mitigate low-frequency oscillations (0.2–3.0 Hz), a power system stabilizer (PSS) was designed, aiming to provide an additional control signal that enabled the suppression of oscillations during faults or any changes in the operating conditions of the system [5, 6].

One of the biggest challenges faced by designers in improving the reliability of the power system is determining the optimal values of the PSS parameters. In this context, several techniques have been mentioned in the literature, including applied mathematics methods [7, 8], eigenvalue assessment [9], and many other modern techniques [10, 11]. Although these methods have been successful in tuning the control system parameters, their drawback is that they take a long time to converge to the optimal solution and require considerable effort in designing the calculation algorithms.

In recent years, artificial intelligence techniques have started to take a place in addressing the problems of power control system design, such as fuzzy logic [12], adaptive fuzzy logic [13], and artificial neural networks (ANNs) [14]. Among these techniques, ANNs have proven their superiority due to their ability to handle complex nonlinear problems and their capability to process applications that involve large amounts of data. Hence, designers have relied on ANNs to design more reliable and robust controllers [15, 16]. On the other hand, proportional-integral-derivative (PID) controllers have been widely used in industrial processes due to their simple design and their robust

performance against a wide range of operating conditions. Experiments have proven that the PID controller is very effective in minimizing the oscillations in power systems [17]. This article introduces a simple robust controller by combining the features of both the proportional-integral (PI) controller and ANNs. The proposed controller in this research is implemented on a single-machine infinite-bus (SMIB) power system. The next section introduces a brief overview of the components and modeling of this system.

2 Power System Modeling

The power system under study consists of one machine linked to an infinite bus that is commonly modeled using a linearized Heffron-Phillips model with six K -constants as shown in Figure 1 [18], where $\Delta\delta$ is associated with the power angle deviation, $\Delta\omega$ with the rotor speed deviation, and ΔP_m represents the input mechanical power. The transient electromagnetic force (EMF) on the q-axis is represented by $\Delta E'_q$, and ΔE_{fd} is the input voltage excitation. The values of T_{do} , D , and H denote the system components that correspond to the open-circuit transient time constant, damping torque coefficient, and inertia constant, respectively. Finally, K_A is the exciter gain constant, and T_A is the exciter time constant.

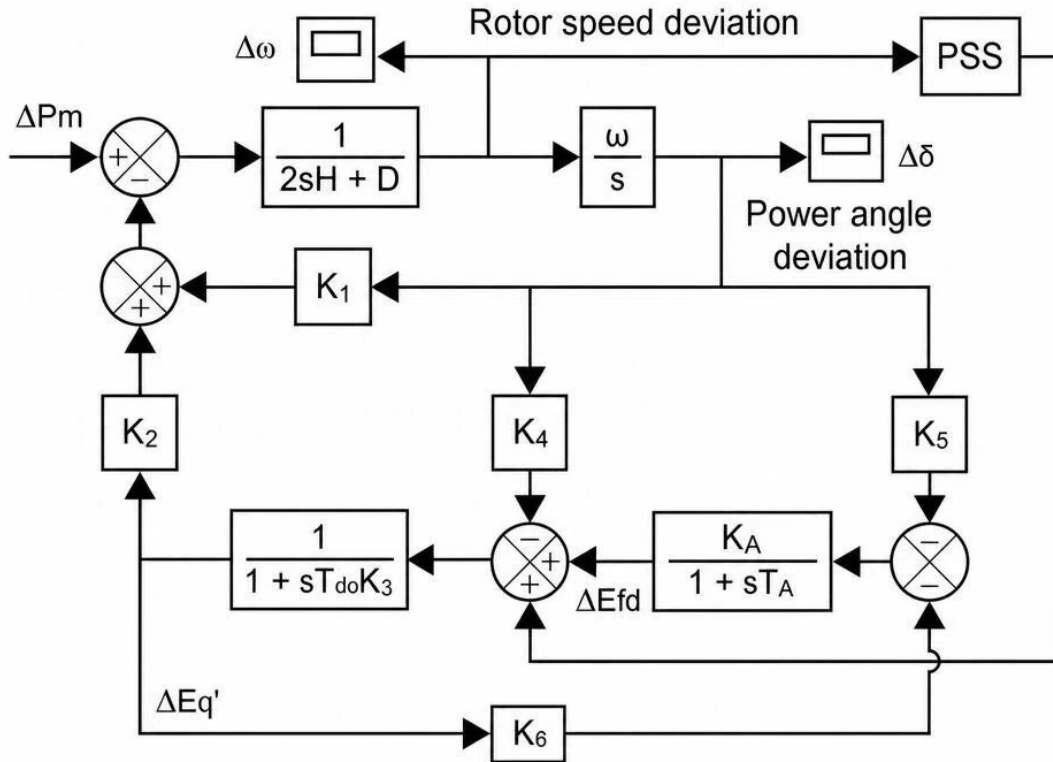


Figure 1. Block diagram of a linearized Heffron-Phillips single-machine infinite-bus (SMIB) with power system stabilizer (PSS)

K -constants of the Heffron-Phillips model are defined as follows: K_1 is synchronization coefficient; K_2 is electric power constant for flux with constant rotor angle, K_3 is impedance factor, K_4 is demagnetizing effect of rotor angle change, K_5 is voltage change in terminal with rotor angle change for constant $\Delta E'_q$, K_6 is voltage change in terminal with $\Delta E'_q$ change for constant rotor angle [18]. It is notable that the values of the K -constants depend on the impedance ratio as well as the loading condition, namely the active and reactive power absorbed by the network, which causes a significant change in the dynamic behavior of the machine at different loading points [19].

3 Neuron-Proportional-Integral Controller

Prakasa and Robandi [20] introduced an equilibrium optimization algorithm (EOA) for tuning the parameters of the PSS that consists of a gain block K_s , a washout block with time constant T_w , and a lead-lag block that has four parameters T_1 , T_2 , T_3 , and T_4 . Such optimization algorithms for driven PSS systems typically seek to tune the high-dimensional parameter space of standard lead-lag networks, which involves adjusting up to six parameters. Simultaneously, this multi-dimensional tuning significantly escalates the computational burden and leads to processing latency during severe dynamic shifts. The proposed NEURON-PI framework introduces a key structural advantage. By using the adaptive capability of the adaptive linear neuron (ADALINE) network in a

compact PI configuration, the number of online updated control parameters is structurally reduced from six to only two, which are the proportional control gain (K_p) and the integral control gain (K_i). This approach avoids the curse of dimensionality, ensures fast learning convergence without tracking delays, and completely eliminates the requirement for offline metaheuristic optimization algorithms like the EOA. Figure 2 shows the general structure of the proposed NEURON-PI controller, where $U(s)$ is the control signal.

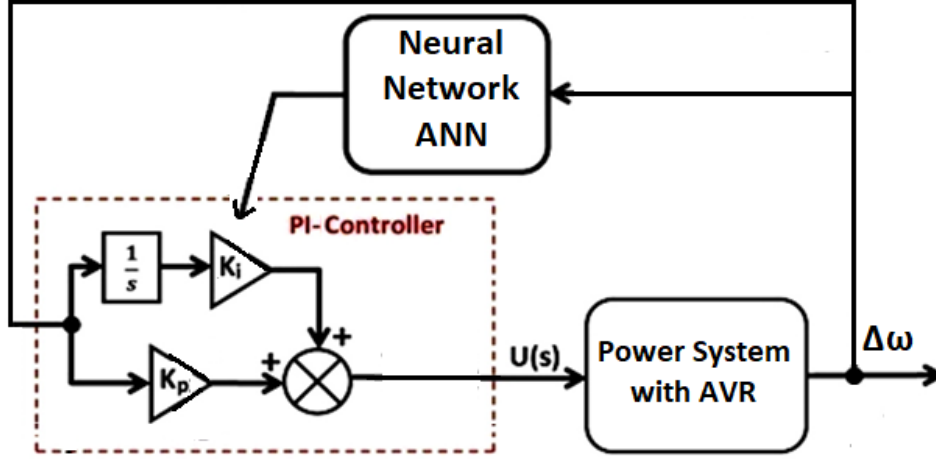


Figure 2. Block diagram of NEURON-PI controller

Note: ANN—artificial neural network; AVR—automatic voltage regulator; NEURON-PI—the proposed adaptive neuron controller.

The ADALINE network can be either multi-input—multi-output or multi-input—single-output. The ADALINE weights are set randomly at the beginning, and then they are updated through the Back Propagation Algorithm by computing the error factor before sending it back to the input units. Finally, the network output is determined according to training input patterns as well as the activation function. The training process mentioned above is repeated to achieve minimum error [21]. The detailed structure of the proposed NEURON-PI controller under the MATLAB/Simulink environment is shown in Figure 3.

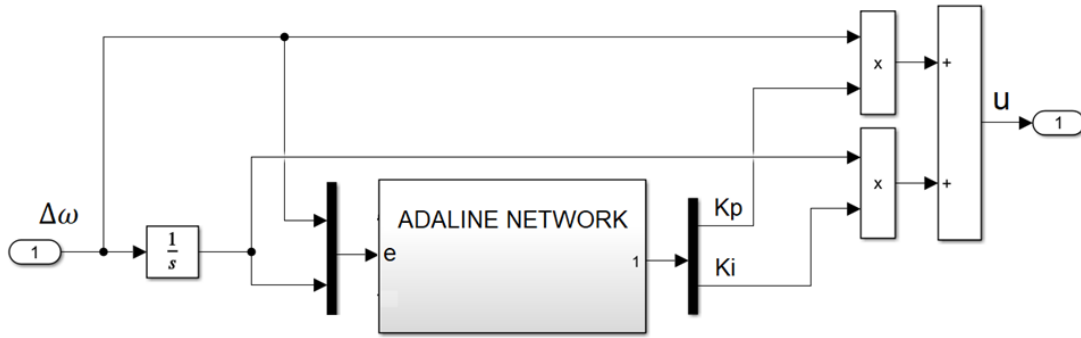


Figure 3. The detailed structure of the proposed NEURON-PI controller environment

Note: NEURON-PI—the proposed adaptive neuron controller.

4 The Adaptive Linear Neuron Learning Algorithm

To ensure the reproducibility of the proposed adaptive control scheme, the mathematical foundation of the ADALINE is explicitly defined. The ADALINE network dynamically tunes the PI controller gains based on the real-time system state. Let the input vector to the ADALINE at an instantaneous time be defined as:

$$X(t) = [x1(t), x2(t), \dots, xn(t)]^T \quad (1)$$

where, the inputs are synthesized from the system error signal, specifically the rotor speed deviation ($\Delta\delta$). Since ADALINE employs a linear activation function, the network output $y(t)$ is directly equal to the linear combiner output, which is the net weighted sum before the activation function, expressed as:

$$y(t) = W^T(t)X(t) + b(t) \quad (2)$$

where, $W(t) = [w1(t), w2(t), \dots, wn(t)]^T$ represents the adjustable weight vector, and $b(t)$ is the bias term. The output $y(t)$ mapping is utilized to compute the dynamic adjustments for the controller parameters (K_p and K_i). The objective of the learning algorithm is to minimize the instantaneous squared error cost function, defined as:

$$J(t) = \frac{1}{2}e(t)^2 \quad (3)$$

where, $e(t)$ is the tracking error performance index. Applying the steepest descent method (Widrow-Hoff Least Mean Squares (LMS) algorithm), the gradient of the cost function with respect to the weight vector is computed as:

$$\nabla_W J(t) = \frac{\partial J(t)}{\partial W(t)} = -e(t)X(t) \quad (4)$$

Consequently, the governing weight-update equation that establishes how the network parameters evolve during online operation is formulated as:

$$W(t+1) = W(t) + \eta e(t)X(t) \quad (5)$$

where, η denotes the learning rate (set to 0.9 in this work). To prevent numerical divergence and eliminate weight drifting during steady-state conditions, a stabilizing factor γ and a bounding limiter L are integrated into the practical implementation, yielding the modified resilient update law:

$$W(t+1) = \text{sat}(W(t)(1 - \gamma\eta) + \eta e(t)X(t), -L, L) \quad (6)$$

where, $\text{sat}()$ represents the standard saturation function bounding the parameter space, defined as:

$$\text{sat}(f, -L, L) = \begin{cases} -L & , \quad f < -L \\ f & , \quad -L \leq f \leq L \\ L & , \quad f > L \end{cases} \quad (7)$$

5 Parameter Selection and Sensitivity Discussion

The credibility and robust convergence of the neural network depend significantly on the choice of its architectural parameters: the learning rate η , the stabilizing factor γ , and the weight limiter L . A qualitative sensitivity analysis is conducted to establish how these selected parameters influence system stability and controller convergence:

1. The learning rate dictates the step size taken towards the minimum of the error surface. A relatively high learning rate ($\eta = 0.9$) is selected to guarantee rapid adaptation of the PI gains (K_p , K_i) when faced with sudden grid disturbances. While an excessively large η can induce severe weight oscillations and potential closed-loop instability, the coupling with the system dynamics in this design ensures rapid, monotonic convergence without dead-time delays;

2. The stabilizing factor is set to be ($\gamma = 10^{-6}$). This parameter acts as a weight-decay mechanism. It provides a small bounded regularization that prevents the weights from shifting unboundedly (weight wind-up) during sustained steady-state operation, thereby guaranteeing bounded-input bounded-output (BIBO) stability;

3. The weight Limiter defines the strict saturation boundaries for the parameter space. It prevents numerical overflow during severe transient states and eliminates actuator saturation, ensuring that the synthesized control signals remain within safe physical operating limits of the exciter system. The weight limiter is selected as ($L = 10^6$).

6 Simulation Results and Discussion

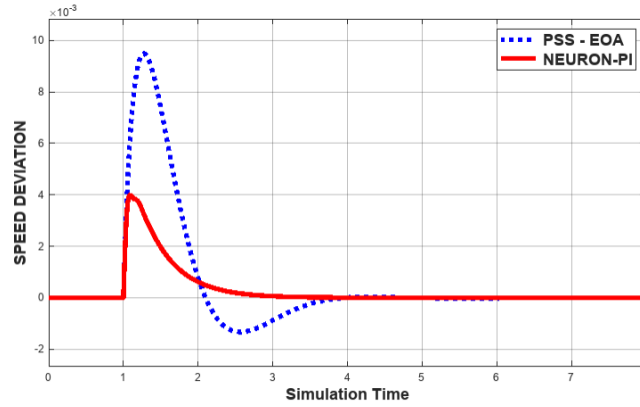
To rigorously evaluate and verify the dynamic performance of the proposed Adaptive NEURON-PI controller, the SMIB system was subjected to a strict load perturbation. Specifically, a step change of 0.02 pu in the mechanical torque (ΔT_m) was applied as a driving disturbance at $t = 1.0$ s. The transient responses of the system, focusing on the rotor speed deviation ($\Delta\omega$) and the power angle deviation ($\Delta\delta$), were monitored across two different operating points, namely, normal load and heavy load. The simulation was conducted for the two types of controllers, including PSS tuned by EOA [20] and the NEURON-PI controller proposed in this article. All parameters of the power system, PSS, and the values of K-constants related to system loading are listed in Table 1:

Table 1. System and power system stabilizer (PSS) parameters

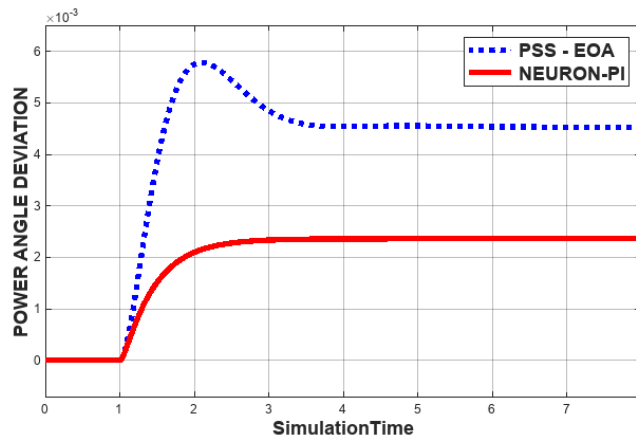
Component	Parameters/Values					
Generator & exciter	Inertia constant (H)	Damping coefficient (D)	Damping coefficient (T_{do})	Electrical angular frequency (ω_e)	AVR gain (K_a)	AVR time constant (T_A)
	3.5	0	7.23	314	200	0.2
Loading point	K_1	K_2	K_3	K_4	K_5	K_6
Normal load	0.760	0.860	0.320	1.420	0.150	0.420
Heavy load	1.635	0.842	0.360	1.077	-0.032	0.521
PSS-EQA [18]	Stabilizer gain (K_s)	Wash-out time constant (T_w)	Lead time constant (T_1)	Lead time constants (T_2)	Lead time constant (T_3)	Lead time constants (T_4)
	9.47	1.53	0.43	0.81	1.01	0.35

Note: PSS-EQA—power system stabilizer-equilibrium optimizer algorithm; AVR—automatic voltage regulator.

The simulation results display the comparisons between the performances of the proposed NEURON-PI and conventional PSS in terms of the rotor speed deviation and power angle deviation of the SMIB system under two case studies. In the first case, the normal load condition was considered, and the superiority of the NEURON-PI controller was clearly notable in a reduction of the system's overshooting and settling times compared to conventional PSS. Figure 4 shows the comparison of rotor speed deviation response with both NEURON-PI and PSS, while the comparison of power angle deviation response is depicted in Figure 5.

**Figure 4.** Rotor speed deviation of single-machine infinite-bus (SMIB) with power system stabilizer (PSS) and NEURON-PI controllers for normal load

Note: PSS-EQA—power system stabilizer-equilibrium optimizer algorithm; NEURON-PI—the proposed adaptive neuron controller.

**Figure 5.** Power angle deviation of single-machine infinite-bus (SMIB) with power system stabilizer (PSS) and NEURON-PI controllers for normal load

Note: PSS-EQA—power system stabilizer-equilibrium optimizer algorithm; NEURON-PI—the proposed adaptive neuron controller.

In the second case, the heavy load condition was considered. It can be seen that the NEURON-PI can reduce the overshoot and the settling time of the system response, and the NEURON-PI controller was faster than the PSS. The deviation of rotor speed and the deviation of power angle of the machine with both controllers are shown in Figure 6 and Figure 7, respectively. The detailed system response of SMIB is given in Table 2.

Table 2. Overshoot and settling time of power system response

Response/Load Condition		Normal Load		Heavy Load	
		PSS-EOA	NEURON-PI	PSS-EOA	NEURON-PI
Speed deviation	Overshoot (pu)	0.0094	0.0039	0.0072	0.0042
	Settling time (s)	3.69	2.02	2.2	0.7
Angle deviation	Overshoot (pu)	0.0013	0.000	0.0052	0.0003
	Settling time (s)	3.3	2.1	2.3	0.8

Note: PSS-EQA—power system stabilizer-equilibrium optimizer algorithm; NEURON-PI—the proposed adaptive neuron controller.

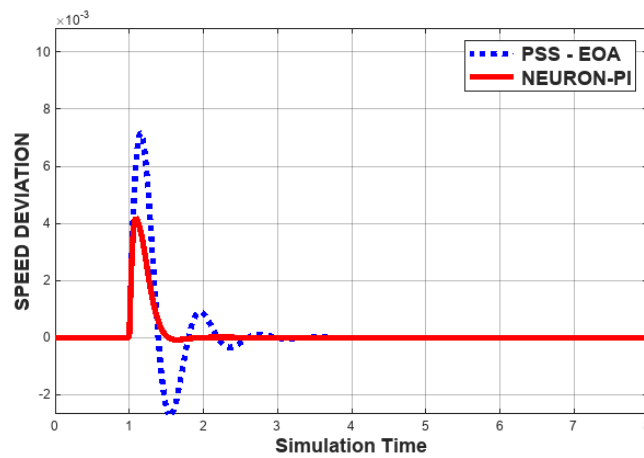


Figure 6. Rotor speed deviation of single-machine infinite-bus (SMIB) with power system stabilizer (PSS) and NEURON-PI controllers for heavy load

Note: PSS-EQA—power system stabilizer-equilibrium optimizer algorithm; NEURON-PI—the proposed adaptive neuron controller.

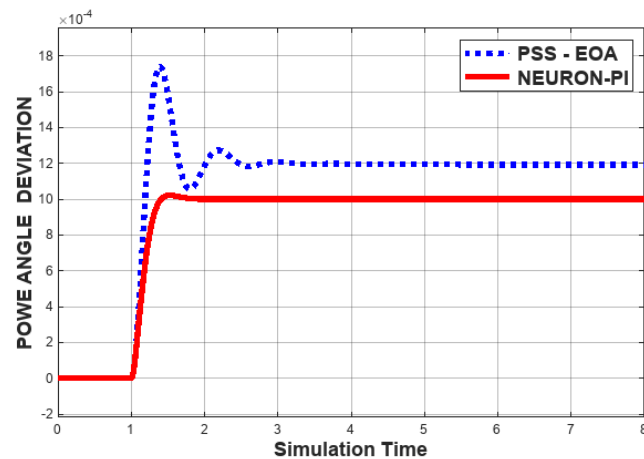


Figure 7. Power angle deviation of single-machine infinite-bus (SMIB) with power system stabilizer (PSS) and NEURON-PI controllers for heavy load

Note: PSS-EQA—power system stabilizer-equilibrium optimizer algorithm; NEURON-PI—the proposed adaptive neuron controller.

7 Conclusion

In this study, an advanced, simplified, and robust control paradigm has been successfully conceptualized and validated for an SMIB power system utilizing a linearized Heffron-Phillips model. The core contribution of this

research lies in replacing the cumbersome conventional PSS with an intelligent, self-adaptive NEURON-PI controller driven by an ADALINE neural network. By harnessing the online self-adaptation capabilities of ANNs, the proposed configuration effectively slashes the number of tunable control parameters from six (in the conventional PSS) to only two. This eliminates the necessity for computationally demanding, offline heuristic optimization algorithms, such as the EOA, thereby providing a highly efficient and easily implementable alternative for power system dynamic stability. The dynamic superiority and robust performance of the proposed NEURON-PI controller were rigorously evaluated under highly contrasting operational paradigms, specifically normal and heavy loading conditions. Quantitative analysis of the simulation results revealed a phenomenal enhancement in the system's transient response. Under normal load conditions, the proposed controller decreased the rotor speed deviation settling time from 3.69 seconds to 2.02 seconds, while completely eliminating the power angle overshoot. More importantly, under critical heavy load conditions, the NEURON-PI controller demonstrated an exceptional ability to damp low-frequency electromechanical oscillations, aggressively compressing the rotor speed settling time by approximately 68% (from 2.2 s down to 0.7 s) and the power angle settling time by 65% (from 2.3 s to 0.8 s). These metrics decisively confirm that the adaptive NEURON-PI design provides vastly superior damping, faster stabilization, and higher structural robustness against severe operational shift points compared to EOA-tuned PSS.

Author Contributions

Conceptualization, I.A.; methodology, A.A. and I.A.; software, I.A.; validation, A.A.; formal analysis, A.A. and I.A.; data curation, A.A.; writing—original draft preparation, I.A. and A.A.; writing—review and editing, A.A. and I.A.; visualization, I.A.; supervision, A.A.; project administration, A.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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